

Poisson regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned}y_i \mid \mu_i &\sim \text{Poisson}(\mu_i) \\ \log(\mu_i) &= \beta_0 + \beta_1 x_i\end{aligned}$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned}\mathbf{y} \mid \boldsymbol{\mu} &\sim \text{Poisson}(\boldsymbol{\mu}) \\ \log(\boldsymbol{\mu}) &= \mathbf{X}\boldsymbol{\beta}\end{aligned}$$

3. Worked observation ($i = 1$)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\underbrace{y_1}_{\text{observed}=1} = 0.955 + -0.0438 \cdot 0.45 + 0.0643 = 1$$

Predicted $\hat{\mu}_1 = 0.936$; residual $\hat{\varepsilon}_1 = 0.0643$.